

The Effect of Stock Spam on Financial Markets

WORKING PAPER

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Abstract. Spam messages are ubiquitous and extensive interdisciplinary research has tried to come up with effective countermeasures. However, little is known about the response to unsolicited e-mail, partly because spammers do not disclose sales figures. This paper correlates incoming spam messages that promote the investment in particular equity securities with financial market data. We use multivariate regression models to measure the impact of stock spam on traded volume and conduct an event study to find effects on market valuation. In both cases we have found evidence for significant reactions to spam campaigns in the short run. Theoretical and practical implications of the findings are addressed.

Keywords: Stock Spam, Event Study, OTC, Unsolicited Bulk E-Mail, Economics of Information Security [JEL G14, D84, M30, C88]

1 Introduction

Unsolicited bulk e-mails (UBE) are messages sent blindly to a very large number of recipients. This phenomenon commonly known as *spam* is increasingly causing problems in communication networks and undermines the usefulness of e-mail as communication medium. Spammers, the individuals who send UBE, often work in secrecy. Therefore little is known about their proceeding, and almost nothing about their success in terms of response pattern and rates.

Spam is an annoying problem for both business and private users of e-mail. A recent study reports that almost 70 % of all e-mail messages received by an average Internet user are spam messages [1]. In typical spam messages, the sender advertises goods and services, e.g., pharmaceutical products, mortgages, or access to certain websites. Besides being an annoyance, this flooding with unsolicited e-mail messages is also an information security problem. It is comparable to Distributed Denial-of-Service (DDoS) attacks that let computer systems or entire networks fail to deliver the intended functions by overloading it with a high number of unnecessary service requests. There exist no effective countermeasures against this sort of attack. The losses caused by spam are also economically significant. The economic costs associated with spam can be broadly separated into three classes, namely waste of bandwidth, waste of storage capacity, and waste of human (employees') time to sort out unsolicited messages [2].

In this paper, we try to shed some light into the question whether and how recipients react to spam messages. We do this by regarding a specific form of spam, namely *stock spam* that advertises equity securities traded on over-the-counter (OTC) markets. This allows us to correlate spam arrival from a number of probe e-mail accounts with publicly available market data and thus draw inference on the effectiveness of UBE.

The paper is structured as follows: In Section 2, we briefly review prior art on the economic reasons for the spam problem, possible countermeasures, as well as empirical work related to our contribution. Section 3 analyzes the effect of stock spam on the stock market. We use multivariate regression models to assess the impact of stock spam on traded volume and an event study method to measure the influence of stock spam on market price developments. We conclude the paper with a discussion on the limitations of our approach and directions for future work (Section 4).

2 Background and Related Work

Spam has a track record in the literature of many areas. Network security mainly studies how spammers operate by taking over hundreds of badly maintained computers to use their bandwidth [3]. Scholars in computer-linguistics and machine learning deal with the construction of efficient filter algorithms [4]. And social scientists try to understand the motivations of spammers and conceive appropriate policy measures to tackle the problem from a legal and economic side. Here we review only the latter aspects in more detail.

2.1 Economics of Spam and Countermeasures

It has been argued many times that spam is largely a problem of economic incentives [5, 2]. The extraordinary small costs per offer placement make it the preferred medium for advertising products on the “long tail” of the demand curve, which cannot be efficiently promoted with traditional means of advertising (see Table 1). As the cost per contact is so low, spammers do not bother about targeted distribution and already very tiny response rates let the business model break even. The resulting inefficiencies due to information overflow have been studied both in formal economic models [6] and in laboratory experiments [7].

Besides technical solutions using filter mechanisms and laws for litigation and deterrence, it has been suggested that increasing the cost of sending a message would solve the problem at its roots. In the absence of a suitable micro-payment system and due to the differences in income among Internet users, Dwork and Noar [9, 10] first suggested in 1992 to use computing cycles as a unit of account. In the so-called “proof-of-work” schemes, the sender of an e-mail must enclose the solution of a unique and computational hard problem, which is verified at the recipient’s mail server before delivery. For legitimate use of e-mail, this computation should not result in unacceptable delay. However, spammers would not be able to send bulk messages since their (finite) computing resources are

Table 1. Cost of offer placement for common approaches

	Total cost	Number of recipients	Cost per recipient
Direct mail	\$ 9,700	7,000	\$ 1.39
Telemarketing	\$ 160	240	\$ 0.66
Print - targeted	\$ 7,500	100,000	\$ 0.075
Print - general	\$ 30,000	442,000	\$ 0.067
Fax	\$ 30	600	\$ 0.05
Online ads	\$ 35	1,000	\$ 0.035
Spam	\$ 250	500,000	\$ 0.0005

Source: [8]

constrained. One possibility to construct such hard-to-solve but easy-to-verify problems uses hash functions and is therefore known as *hashcash* [11]. Laurie and Clayton [12] criticize these proposals for two reasons. First, the additional problem-solving burden would also affect legitimate users to a non-negligible extent. Second, spammers access insecure end-user machines to steal processing cycles and solve puzzles. Instead they suggest using CAPTCHAs [13], a class of proof-of-work puzzles that requires human interaction, which is presumably more difficult to “steal”. Other approaches target in similar directions, such as Loder et al. [6], who propose a scheme in which the recipient of a message can decide whether or not to charge the sender, and Fahlman [14], who suggests making attention to a tradable good by allocating “interrupt rights”. It is up to see in the future whether such schemes can result in socially optimal outcomes.

2.2 The Stock Spam Business Model

The general proceeding of spammers and the underlying business model is simple. Spammers act rationally and try to maximize their (risk-adjusted) expected profit, similar to all other types of economic agents. In contrast to other sorts of *sales spam*, stock spammers do not directly offer a product or service. They rather speculate on positive price developments of thinly traded stocks after they have been hyped in thousands of messages sent to possible investors. The content of such spam messages often pretends to be a misdirected investment advice, enriched with financial terms and recent price quotes. Especially in low liquidity markets with few information coverage, the mere attention of a particular stock may stimulate an investment decision [15]. If one believes that many people follow such dubious “investment advices” then jumping on the bandwagon is not irrational, since virtually everybody could profit from speculative gains in the resulting bubble. The persistence of such spam, as well as the results presented below, let us conclude that this pump-and-dump strategy actually works.

It might even work so well that “e-mail marketing” of stocks is openly offered on the Internet. For example, Expedite [16] claims that

“[...] e-mail marketing .com is a full service OTC Pink Sheet Stocks e-mail marketing company that can e-mail out your OTC stocks newsletter to the masses. [...] With our stable and reliable network and bandwidth, we can service any size of OTC Pink Sheet stock awareness campaign.”

Our analysis below will show how the masses react ...

2.3 Stock Spam Watchers

Stock spam has been discussed so far on a number of blogs, and some websites collect information on stock spam information. Cyr runs a *Spam Stock Tracker* [17] since March 2005, where he keeps track of the performance of securities that have been advertised in spam messages. For each unique stock, he adds 1,000 shares to a fictive portfolio. As of March 15th, 2006, he (virtually) suffered a net loss of US\$ 27,827 bar transaction costs. This shows that the long-term performance of advertised stocks has been negative on average. In contrast to this long-term analysis, Richardson’s *Stock Spam Effectiveness Monitor* [18] provides a graphical summary of the intra-day development of advertised stocks. Finally, the web source [19] lists an (incomplete) collection of affected firms together with example messages, and McIntyre [20] requests and collects comments from firms that were cited in stock spam messages. Hence, to the best of our knowledge, this paper seems to be the first academic study dealing with stock spam.

2.4 Related Event Studies

Later in this paper we will use the *event study* methodology to empirically measure the influence of stock spam dissemination on the market price development of the affected stocks. This method is a standard approach that has been applied to numerous research questions in finance and economics [21]. The method is also not novel in the context of computer security. Several authors have investigated the impact of public security incident reports on the stock market valuation of affected firms [22–24] and software vendors [25]. All studies consistently report a negative and significant market impact. The event study methodology has also been applied in analyses of “serious” investment advice (unlike stock spam), however with varying results. In [26] the independent variable is constructed from recommendations of financial analysts, whereas the authors of [27] use recommendations printed in the mass media as predictor for stock price development. We are not aware of a paper that discusses particularities of the event study methodology for small- and micro-caps, the type of stocks we regard in our analysis.

3 Stock Market Impact of Unsolicited E-Mail

The empirical work described in this section is the core of our contribution. We start with a presentation of the data source (3.1), then continue with descriptive analyses of stock spam activity (3.2) before we analyze the impact of stock spam arrival on traded volume (3.3) and market valuation (3.4). As the methodology differs between variables of interest, we discuss it in the respective sections.

3.1 Data Acquisition

Our empirical study is based on the following data sources. The spam events were downloaded from Richardson’s *Stock Spam Effectiveness Monitor* (SSEM)

archive [18]. The data comprises 21,935 stock spam messages between November 2004 and February 2006. The messages were extracted automatically from a number of spam collecting e-mail addresses. On average, 3 % of all incoming messages were classified as stock spam [18]. The corpus of spam messages cites 391 unique stocks, which corresponds to about 5 % of all stocks listed on the relevant OTC markets: 68 % of the stocks in our sample are listed on the National Quotation Bureau’s (NBC) *Pink Sheets*, a financial services company distributing real-time price information on over-the-counter transactions of penny stocks. The remaining part refers to stocks quoted on the *OTC bulletin board* (OTCBB), a similar entity for public firms that fulfill some financial reporting requirements but still do not meet the rigorous listing standards of the major U. S. exchanges [28]. We believe that stock spam exclusively targets small- and micro-cap securities (so-called penny stocks) because the spammers bargain for a positive market impact due to their activity. Market impact, i.e., the reaction of the market price on individual orders, is generally higher for low liquidity securities. To assess the validity of this data source we compared some of the stock spam messages in the authors’ personal e-mail accounts to SSEM data and found a relatively good correspondence with respect to the stocks cited on specific days.¹

Daily price quotes for the affected tickers² were downloaded from *Yahoo Finance* [29]. Unfortunately, no historical data was available for a number of tickers. Therefore the usable data set was reduced to 111 (28.4 %) tickers and 7606 (34.7 %) relevant spam messages. There is no obvious reason to suspect that this selection systematically affects the results due to a coverage error between the stocks where data is available in Yahoo Finance and those where it is not. Future research can improve validity by acquiring more complete financial data.

To assess the contribution of a market model in the event study [21], we selected three daily market indices: Standard & Poor’s 500 and NASDAQ Composite were both obtained from Yahoo Finance. They are very common indicators for general stock market performance in the U. S., but both are computed from high liquid securities only. Therefore we decided to include Russell’s daily micro-cap index as well. Its historical data (until December 2005) has been downloaded directly from the data provider’s website [30].

3.2 Descriptive Data Analysis

Aggregating the SSME data allows to construct a good indicator for stock spam activity over time. The solid line in Figure 1 displays a smoothed time series of the total number of stock spam messages received on the collecting addresses. The absolute figure is not particularly informative since it depends on the number of probe accounts. However, it is reasonable to assume that the total number of spam messages distributed varies proportional to this indicator. Note that November 2004 and February 2006 are not completely represented in the data, so that mainly the course of 2005 should be regarded as core period of interest.

¹ We never experienced identical messages as spammers apparently vary message subjects and pretended sender names systematically to elude simple spam filters.

² A *ticker* symbol is a unique identifier for traded stocks.

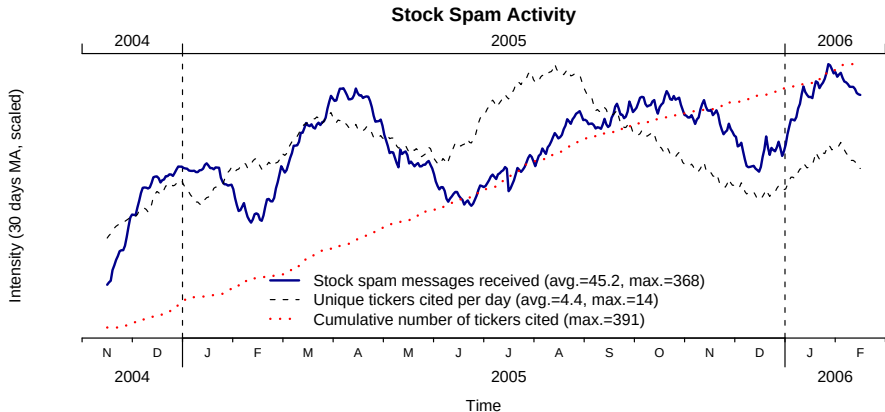


Fig. 1. Time series of total stock spam messages in the data set ($n = 21,935$). Joint graph of a) 30-day moving average of daily message arrivals (solid line), b) 30-day moving average number of different tickers cited in one day’s total spam (dashed line), and c) cumulative number of affected companies over time (dotted line). All series are scaled to a unit interval. Only a small subset of these events is included in the multivariate analysis.

We are not aware of examples where more than one ticker is mentioned per spam message, but for the majority of days the data contains references to a number of different tickers in separate messages. Therefore the dashed line shows the development of the number of unique ticker symbols being cited in the total stock spam of each day. It would be too far-fetched to interpret this as a sign of competition between spammers, but it is also difficult to imagine how this “diversity” could be planned to support one single spammer’s strategy. Imagine it were a sign of competition, then we could interpret the dynamics between number of unique tickers and the number of messages as a decline in competition from August 2005 onwards. In other words, spammers concentrate again on fewer tickers per day after they drove the number up to 14 in August 2005 (here the absolute numbers make sense if we believe that the data does not systematically miss large parts of stock spam traffic).

The dotted line in Figure 1 shows the cumulative number of tickers being cited in stock spam from the beginning of the data set. It tells us that constantly new firms become victims of stock spammers. At the same time, some stocks remain targets of spam attacks for quite a long time and thus accumulate an impressive number of messages distributed over up to 77 event days. See Tables 6 and 7 in the appendix for a ranking of the most seriously hit tickers by number of events and total messages, respectively.

Figure 2 breaks the message arrival further down by weekdays and daytime. It is clearly visible that the large majority of messages arrives on working days,

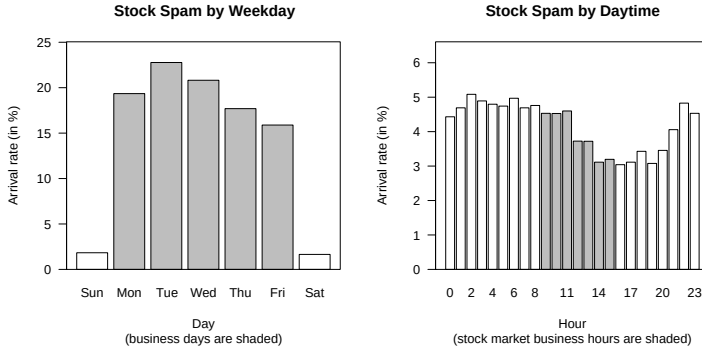


Fig. 2. Distribution of stock spam message arrivals across weekdays (left) and the course of a day (right, U.S. eastern time). Spammers apparently avoid weekends but do not bother a lot about market hours. In the analysis, messages received after the close of the market are counted as events on the following business day (*effective day*).

although Sunday afternoon arrivals (after 4:00 p.m.) were already counted to the Monday numbers. This is due to the processing logic that assigns message arrivals to business days, which is automatically performed at the data collection stage: as the Pink Sheets and OTCBB follow regular market hours, from 9:30 a.m. to 4:00 p.m. US eastern time [31], all messages received after the market had been closed were moved to the next business day. Therefore the *effective day* in our study does not necessarily match the actual calendar day of message arrival. In case of weekends and business holidays, we additionally shift the effective arrival time by 24 hours (but not more than three times in a row).

Unless otherwise stated, we will further use the term *event* to express the arrival of one or more messages citing a particular ticker on a specific (effective) day. By contrast, we use the term *quantity* in those parts of the analysis where the actual number of messages per day citing the same stock is a relevant measure.

3.3 Effects on Traded Volume

If stock spam actually has an influence on the markets then it should most easily be seen in the trading activity. Stock spammers exclusively target penny stocks, presumably because the market impact of individual transactions is particularly high for securities with low liquidity. In most cases, the liquidity is so low that there are business days where a penny stock is not traded at all. Therefore, the simplest way to test the impact of stock spam is a cross-tabulation of trade activity and spam arrival, as shown in Table 2. In fact, we see a positive relationship which is also statistically significant using Pearson’s χ^2 statistic for contingency tables.

Though its message is very clear, this test is certainly too simple to provide sound evidence for a positive relationship, because a number of possible third

Table 2. Effect of spam arrival on trade activity (per business day)

Trade volume	Stock spam received	
	No	Yes
= 0	15.8 %	2.7 %
> 0	84.2 %	97.3 %
	100.0 %	100.0 %
	($n = 32261$)	($n = 547$)
$\chi^2(1) = 68.5, p < 0.001$		

variables are not controlled for. Hence, we turn away from the binary response case (trade / no trade) to a quantitative evaluation of the impact of spam arrival on the traded volume. The graphs in Figure 3 visualize the differences in average volume per stock on a linear (left) and log (right) scale. All 111 stocks in the sample are sorted by their average volume at normal days. The large range of average volumes illustrates the heterogeneous composition of our sample.

Form visual inspection one might already assume a tiny positive influence of stock spam in both graphs. Multivariate regression models are the right tool to quantify this relationship and test the hypothesis on data. Due to the varying daily turnover between stocks, we opt for a multiplicative model formulation, where the average volume on days with spam arrival can be expressed as a product of the individual stock’s average volume on normal days times a “spam impact factor” α . As a result, however, we have to exclude cases without trade since any volume increment above zero would result in infinitely high factors α and thus render the regression problem intractable or yield misleading results as artifacts of possible correction measures (such as replacing zeros by very small nonzero values). This is our baseline model M1:

$$v_{t,i} = v_0 \cdot e^{\zeta_i} \cdot w_{(t)} \cdot \beta_0^{\lambda_t} \cdot \alpha^{\delta_1(x_{t,i})} \quad (1)$$

In our notation, $v_{t,i}$ is the (strictly positive) trade volume of stock i at day t . v_0 is the average volume, and ζ_i is a stock-specific scaling factor for the overall volume, where we assume $\zeta_i \sim N(0, \sigma_\zeta^2)$. ζ_i actually models the heterogeneity between stocks.³ To control for possible influences of time, we include $w_{(t)}$, a vector of four coefficients to capture variations in volume between days of the week, and λ_t , a rational scaled time variable ranging between 0 and 1 from the first day to the last day of the sample period (478 days in total). Function $\delta_1(\cdot)$ converts the absolute number of spam messages $x_{t,i}$ received at day t and citing

³ Readers who deem the normality assumption in the random-effects model as too strong should note that we have tested alternative models with 111 fixed effects, one per stock. The estimates for $\log(\alpha)$ tallied up to 2 digits behind the decimal point.

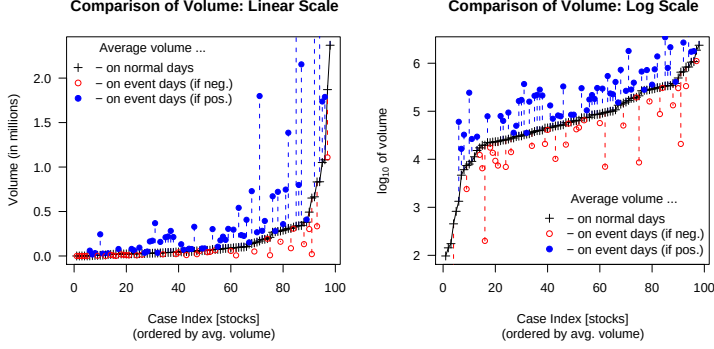


Fig. 3. Visual analysis of average daily trading volume per stock on normal days (smooth line with cross markers) and event days with at least one stock spam message received (buzzing points) both on linear (left) and log (right) scale. Differences are plotted as dashed lines.

stock i to a binary dummy variable:

$$\delta_1(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Log-linearization of Eq. 1 yields a linear regression model with random effects term that can be fitted to data using restricted log-likelihood maximization (REML) to estimate the spam impact on volume as parameter α [32].

$$\log v_{t,i} = \log v_0 + \zeta_i + \log w_{(t)} + \log \beta_0 \cdot \lambda_t + \log \alpha \cdot \delta_1(x_{t,i}) + \epsilon_{t,i} \quad (3)$$

The estimated coefficients are reported in column M1 of Table 8 in the appendix. As $\log(\alpha)$ is positive and highly significant, we found evidence for the presence of a relationship between spam events and the amount of stocks traded. As to the controls, there is only negligible influence from weekdays (all $w_{(t)}$ do not significantly differ from zero) and we capture a positive linear trend in the traded volume of our sample of stocks ($\beta_0 > 0$), which might be a concomitant of the upswing position in the business cycle.

The actual value of α allows us to compute the average change in volume of a stock on days with message arrival compared to normal days, where the ticker has not been cited in stock spam. As displayed in Table 3, the impact is quite high: spam events make volume more than triple.

However, this relationship does not yet support the conclusion that the additional volume is actually caused by the recipients of stock spam messages. It is also possible that the senders commit large parts of the transactions through buying stocks before spamming and selling (at a higher price if the business work) after the market has reacted. Moreover, the relationship could also stem from an inverse causality, namely when the spammer pursues a strategy to select

Table 3. Effect of spam arrival on trading volume

Model	Avg. volume reaction on spam event	95 % confidence interval	No. of events
All spam events	+215.2 %	176.2–259.7 %	532
Spam before market hours only	+154.1 %	107.9–210.6 %	222

particularly those stocks as targets that show exceptionally high volumes.⁴ To exclude at least this last hypothesis of inverse causality, we re-estimated model M1 on a sub-sample by dropping all events where messages have been received during market hours. Hence, the spammer could not have had known the volume at the time the message was sent. The results, as reported in the second row of Table 3, indicate a somewhat lower but still big and highly significant effect. Note that some reduction is expected since now about half of the spam days’ high volumes account to the average of normal days. Consequently, the constant term of M2 is slightly higher than for M1 (see Table 8 in the appendix). We conclude that spammers probably do not select their targets by reacting to high volumes at the same day, and continue our analyses with the full set of events.

In model M3, we further relax the assumption that a spam event is a binary state and estimate the relationship between the message quantity, in terms of messages received per day, and trading activity. In absence of a reasonable prior for the functional form for the relationship we group the outcomes of cumulative spam arrival $x_{t,i}$ into 8 disjoint bins with approximately equal frequency. Quadratically increasing bin breaks turned out to achieve this goal very well. The model equation is a direct generalization of model M1, replacing one single α by a vector α_k with one element per (nonzero) bin:

$$v_{t,i} = v_0 \cdot e^{\zeta_i} \cdot w_{(t)} \cdot \beta_0^{\lambda_t} \cdot \prod_k \alpha_k^{\delta_2(x_{t,i},k)} \quad (4)$$

$$\log v_{t,i} = \log v_0 + \zeta_i + \log w_{(t)} + \log \beta_0 \cdot \lambda_t + \sum_k \log \alpha_k \cdot \delta_2(x_{t,i},k) \quad (5)$$

Function $\delta_2(\cdot, \cdot)$ maps the actual number of spam messages $x_{t,i}$ citing ticker i at day t to one of 7 disjoint intervals $\{1, 2, [3, 4], [5, 8], [9, 16], [17, 32], [33, +\infty]\}$. Its value is 1 if interval selector k matches the interval of $x_{t,i}$ and 0 otherwise. The estimated coefficients α_k are all positive and highly significant, whereas their absolute value grows – as expected – with the number of messages received. Therefore our positive results in the previous tests are certainly not artifacts

⁴ It is quite likely that spammers do use market information when selecting their targets, since the majority of messages cites current quotes. If the access to real-time data is once in place it can easily be used for additional purposes.

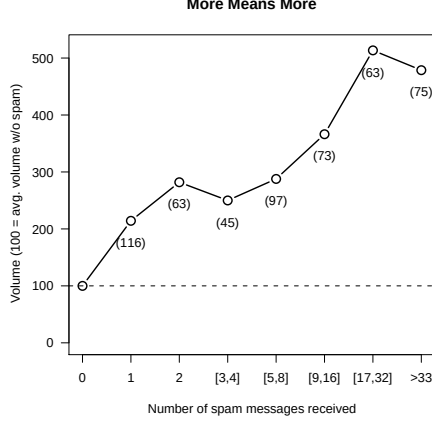


Fig. 4. Effect of the quantity of received messages on traded volume per business day as given by the coefficients α_k of model M3. Categories on the x-axis are quadratically increasing bins. A clearly linear relationship between volume reaction and bin index suggests the existence of diminishing marginal response of additional spam dissemination. Figures in brackets denote number of cases in each bin.

of singular cases with extremely high penetration of spam messages (up to 118 citing the same ticker on a single day). Moreover, a graphical analysis of the estimated impact factors by bins reveals a good linear relationship between bin number and impact (see Figure 4). As bin widths grow quadratically, we find that the spammer faces diminishing marginal “utility” from additional messages. Further developing this admittedly somewhat crazy line of thought, one could come up with an “optimal spam amount” and – assuming that spammers act rationally and operate at that point – eventually infer their implied cost of sending a message (see [5] and [12] for alternative ways to estimate the cost to send spam).

To complete the analysis of effects on volume, we look at the development of effect strength over time. Therefore we specify model M4 as

$$v_{t,i} = v_0 \cdot w_{(t)} \cdot e^{\zeta_i} \cdot \beta_0^{\lambda_t} \cdot (\alpha \beta_1^{\lambda_t})^{\delta_1(x_{t,i})} \quad . \quad (6)$$

The parameters of M4 were estimated from a log-linearized form of Eq. 6, yielding a model with interaction term. The results show positive values for both β_0 and β_1 , whereas only β_0 is statistically significant (see Table 8 in the appendix). This means that the average traded volume of stocks in the sample grew over time, but the effect of stock spam on volume has remained constant (with a slight tendency to the upside). Hence, there is no sign in the data that the “stock spam trick” is wearing out over time.

Table 4. Effect of spam arrival on intra-day stock price development

Intra-day movement	Stock spam received	
	No	Yes
Open>Close	27.8 %	51.9 %
Open=Close	47.1 %	24.3 %
Open<Close	25.1 %	23.8 %
	100.0 %	100.0 %
	($n = 32261$)	($n = 547$)

$\chi^2(2) = 171.5, p < 0.001$

3.4 Effects on Market Valuation

To start with a simple (and naïve) way to assess the effect of stock spam on market valuation of cited stocks, we tabulate the intra-day price development for days with and without spam arrival (Table 4). We find a significant relationship, which again shows that spam actually influences trading activity: the large share of equal open and close prices on days *without* spam reduces by about 50 % for days *with* spam messages. Moreover, the probability mass moves to the cases where the open price is higher than the close price, i.e., where the respective stock loses value. However, considering this analysis as evidence for negative impact in general would be premature for three reasons: *First*, the tabulation approach solely regards the sign and does not take into account the absolute value of profits and losses. If losses are frequent but systematically smaller than (less frequent) profits then the average outcome could still be positive. *Second*, the tabulation includes all spam events (defined as days with nonzero spam arrival rate) irrespectively of possible arrivals in the past. The interactions of effects from subsequent events can be very complex and may bias the result. The *third* concern addresses the fact that the medium-term price development is completely disregarded in this analysis. If a stock price has declined for several consecutive days then even a relatively smaller, but still negative, development at the event day should be regarded as a *positive* effect of spam arrival, and vice versa.

3.4.1 Event Study Methodology

Event study analysis is a technique borrowed from finance research that allows to compensate for the above mentioned shortcomings (for an overview see [21]). The method defines the notion of *abnormal returns* $AR_{t,i}$, that is the difference between the actual daily return $R_{t,i}$ of stock i and its most *normal returns*, i.e., the most likely returns if the event would not have happened $E(R_{t,i}|\theta_i)$,

conditional to a specific prediction model with parameters θ_i :

$$AR_{t,i} = R_{t,i} - E(R_{t,i}|\theta_i) \quad (7)$$

Daily return $R_{t,i}$ are computed from daily adjusted close prices $P_{t,i}$ as follows:

$$R_{t,i} = \frac{P_{t,i} - P_{t-1,i}}{P_{t-1,i}} = \frac{P_{t,i}}{P_{t-1,i}} - 1 \quad (8)$$

Consequently, *cumulative abnormal returns* ($CAR_{t,i}$) are defined as the sum of abnormal returns of a number of subsequent days after an event at time t_0 .

$$CAR_{t,i} = \sum_{\tau=t_0}^t AR_{\tau,i} \quad (9)$$

The crux of the method lies in the choice of a good prediction model to obtain the most likely returns. In [21], MacKinlay discusses constant mean return models, market models, “other statistical models”, and economic models as possible options. All models have in common that their parameters θ are estimated from the stock price development in a time period of size Δt_{est} before the event had been observed. This period is called *estimation window* in contrast to the *event window* of size Δt_{evt} starting at the event day t_0 . An optional gap between the estimation window and the event window can be used to account for possible inaccuracies of event times in the data. See Figure 5 for an illustration of the different time periods in our event study.

Fig. 5. Time line of the event study



We have investigated the prospect to use a market model as predictor. Models of this type predict the expected return conditional to the general market development as measured by a (weighted) index of the stocks under study, or by a common stock market index. The advantage in explanatory power of a market model over the simple constant return model, however, strongly depends on the correlation of individual stocks with the market index or between pairwise stocks in the sample. Therefore we computed the correlation of returns from each stock in the sample to the returns of three popular indices (S&P 500, NASDAQ Composite, and Russell’s specific micro-capital index). The distribution of correlation coefficients is depicted as violin plot in Figure 6 using a kernel density smoother.

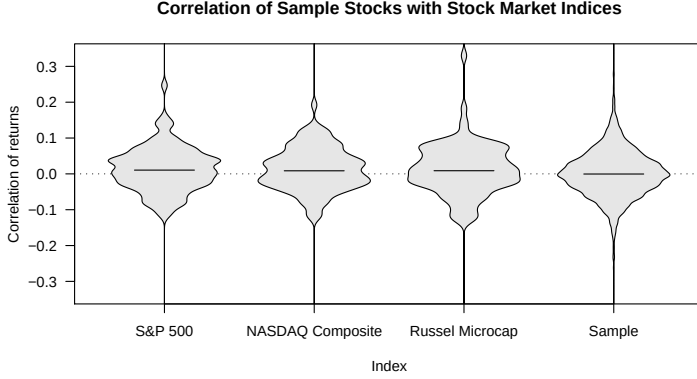


Fig. 6. Violin plot of the density of pairwise correlation coefficients between daily returns of penny stocks in the sample and returns of common market indices (left 3 violins) and other stocks in the sample (rightmost violin). The respective median correlation coefficient is indicated as horizontal bar.

As can be seen from both the distributions and the indicated median correlation coefficients, none of the indices has sufficient explanatory power to make a prediction of returns for the penny stocks in our sample more reliable. This finding is consistent with prior comparisons of (asset-weighted) returns from OTC markets with leading indices [28]. The even lower correlation coefficients between pairs of stocks in the sample thwart any attempt to construct a prediction model from our own sample-specific index. As a result, we employ the constant mean return model for our analysis, where the normal returns are expected to follow the linear trend observed in the estimation window.

$$\mathbb{E}(R_{t,i}|\theta_i) = \theta_i = \frac{1}{\Delta t_{\text{est}}} \sum_{\tau=0}^{\Delta t_{\text{est}}} R_{\tau,i} \quad (10)$$

Although the constant mean return model is the simplest option, finance scholars report that it often yields results similar to those of more sophisticated models [33, 21].

However, the small correlation between returns of stocks in our sample has also a positive aspect for our study: it reduces the risk of intractable interaction effects between overlapping event windows for different stocks, a problem that has been identified in other event studies before and which is usually referred to as “clustering” [21]. Though we do not expect large influences from concurring spam messages citing *different* stocks, we surely must carefully exclude possible influences from subsequent spam events targeting the *same* stock. Otherwise abnormal price developments from previous events could adversely affect the model calibration in the estimation window and, even worse, overlay the dynamics in the event window if multiple events occur in quick succession. Therefore we de-

Table 5. Effect of stock spam on market valuation of cited firms

Day	Aggregated daily abnormal return	95 % confidence interval	Aggregated cumulative abnormal return	95 % confidence interval
0	+1.7 %	[+1.5, +2.0]	+1.7 %	[+1.5, +2.0]
1	−0.9 %	[−1.1, −0.6]	+0.9 %	[+0.5, +1.2]
2	+0.9 %	[+0.6, +1.1]	+1.2 %	[+0.8, +1.7]
3	−1.1 %	[−1.3, −0.8]	+0.2 %	[−0.3, +0.7]
4	−0.9 %	[−1.2, −0.6]	−0.7 %	[−1.3, −0.1]

Constant mean return model on final sample of 152 events for 93 unique stocks.

fine a *penalty period* Δt_{pen} . All events of the same stock with distance less or equal Δt_{pen} are excluded from the analysis (but still act as penalizing entities for subsequent events).

Our analysis was conducted with the following settings: the size of the estimation window was fixed to $\Delta t_{\text{est}} = 30$ business days. An event was excluded if less than two thirds of the required observations were available. We further set $\Delta t_{\text{gap}} = 3$ business days and $\Delta t_{\text{pen}} = 10$ calendar days (the rationale here was that the recognition of spam messages fades regardless of whether the market was opened or closed). This decision dropped 374 events, keeping a total number of 152 (note that multiple message arrivals citing the same ticker at the same day count as one event only). After computing the abnormal returns and cumulative abnormal returns for each event along an event window of size $\Delta t_{\text{evt}} = 5$, four more events were excluded as outliers because their *AR* values for day t_0 were outside five times the 10–90 % percentile of the distribution.⁵

As a sort of robustness check, we tried different alternative settings, which caused the results to change somewhat, but the main effect never disappeared except when the exclusion of subsequent events is completely omitted. Longer estimation windows do reduce the standard errors of the estimates (using the asymptotic distributions as described in [21]) but do not change the general results. At the same time, the number of events decreases because some of the time series for price quotes do not date back long enough before the first relevant events in our sample. Unlike the approach in [21], the statistical hypothesis tests were computed as *t*-tests to better account for the uncertainty linked with a small number of events. All *p*-values reported are more critical than the respective results of the standard method (we verified this for all results). Both methods should concur in the limit case of infinitely many events.

⁵ 3 out of 4 outliers showed extremely high abnormal returns, hence excluding them supports the Null hypothesis and makes our method more prudent.

3.4.2 Results

The main results are summarized in Table 5.⁶ We find a pattern of positive and significant abnormal returns for days t_0 (the effective event day) and t_2 , as well as significant negative abnormal returns for days t_1 , t_3 , and t_4 . This is perhaps an artifact of the interleaved buy- and sell orders from naïve victims (who believe in the message and buy), spammers (who sell after they deem the market reaction is on its maximum), and smart “victims” who jump on the bandwagon and try to buy early to sell soon and thus profit from the price hike as well. But the evidence for this interpretation is not very strong, though a similar pattern remains stable for different subsets of events and also for different settings of the key parameters (not reported here). Regarding the cumulative aggregated returns, the picture becomes much clearer. There exists a non-negligible positive bulge directly after the event that fades over time and finally turns negative on day t_4 (all estimates except t_3 being statistically significant).

The finding of positive abnormal returns is consistent with the presumption that stock spam is on average a profitable business model (for spammers). Otherwise the persistent presence of stock spam would cause a puzzle. However, this finding does not imply that more than 50 % of the cases yield positive returns, nor that the average return is positive. Similar to the initial tabulation in Table 4, a sign test on the individual *AR* estimates (before aggregation but after applying the penalty rule) suggests the contrary: negative and positive abnormal returns occur almost equally frequent. However, the positive outcomes tend to be higher than the negative ones. But even positive abnormal returns do not always imply profits. If the stock is in a downward movement that temporarily looses pace due to spam events, then it is still difficult to make money in this situation. Let alone that spammers quite likely have to bear the transaction costs as well, which are completely disregarded in this study. (N.B. the volume analysis alone suggests that a collusion between market makers charging transaction costs and stock spammers would probably be profitable).

As in the analysis of volume, we have broken down the set of events into smaller subsets to shed more light in interesting subordinated research questions. Similar to model M2 in Section 3.3, we exclude all events triggered by messages that arrived after the market had opened. This allows us to rule out an alternative explanation for the results, namely that spammers deliberately select stocks after they have noticed an exceptional hike in its price. The result pattern of *AR* is very similar to the baseline model; its positive abnormal returns are even slightly higher, which yields more slowly decaying cumulative abnormal returns (see Table 9 in the appendix).

Another break-down has been conducted by liquidity. One might assume that lower liquidity implies higher market impact and thus more favorable (i.e., higher and more predictable) abnormal returns. This rationale also constitutes the common argument why stock spammers target penny stocks only. As we have no direct measure of liquidity in the data, we construct an indicator for liquidity by measuring the fraction of days where high and low prices are equal.

⁶ For more details see Table 9 in the appendix.

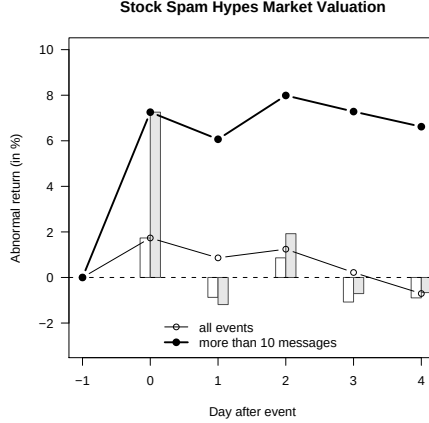


Fig. 7. Effect of the quantity of received messages on abnormal return. Bars indicate aggregated abnormal returns (AR) on single days; line markers show cumulative abnormal returns (CAR). Estimates based on 152 resp. 33 events.

We assume that on those days no transaction has taken place. Though this is only a rough measure – especially for penny stocks where the quantization to ticks may underestimate the actual number of days with transactions – we use this indicator to separate the lower (0–30 %) percentiles from the higher (70–100 %) ones in the sample. Interestingly – and somewhat counter-intuitively – we find that stocks with higher liquidity yield higher and persistently positive cumulative abnormal returns, whereas stocks in the low liquidity group show on average smaller and statistically insignificant reactions at day t_0 , which is quickly driven to the negative in the following days. Hence, the lower liquidity might cause stronger market reactions when spammers and speculators dump their stocks to realize short-term profits. It is up to further research to verify this finding with more appropriate indicators, and to finally decide whether the presumption is right or wrong (then, however, there must be other reasons why spammers usually do not target listed stocks).

We have also analyzed the data for possible aggregated effects of learning over time. Therefore we split the sample into two halves, with July 1st, 2005 marking the break. We observe clearly positive cumulative abnormal returns from the very first (event) day on in the second half only. In the first half, the abnormal return at the event day is not significant, and the measurable reaction is overwhelmingly negative. This result can well be an artifact of the general market conditions, which we did not control for. But it is also possible that traders among the spam recipients made the experience that reacting to stock spam is profitable for them as well, which causes them to jump on the bandwagon and support the self-fulfilling prophecy even further. This interpretation is also

consistent with the positive (albeit not significant) interaction term for the effect on volume over time in model M4 (Table 8).

Finally we looked at the influence of message quantity by calculating the abnormal returns for the subset of events with more than 10 messages per day (citing the same ticker). Here it is clearly visible that more messages drive the abnormal return further to the positive. A visual comparison of the cumulative abnormal returns of the baseline model and the subset with high message quantity is depicted in Figure 7. This is again consistent with our findings on quantity and traded volume.

4 Conclusion

4.1 Discussion

Among all classes of unsolicited bulk e-mail, stock spam has the particular property that reactions can be observed indirectly from publicly available stock price quotes. Our study makes use of this feature and finds evidence that spam message campaigns on average go along with a) an increase in trading activity of the cited stock, and b) positive cumulative abnormal returns shortly after the messages have been distributed. Hence, we conclude that the business model for stock spam actually works. The dynamics in traded volume and prices can be attributed to the sum of actions from at least three groups of individuals:

1. spammers, who trade the stock to capitalize profits from their campaign,
2. naïve recipients, who believe in the pretended investment advises, and
3. smart recipients, who try to participate in price hikes triggered by spammers.

With the given data we cannot disentangle the contributions from each channel. Especially the third group (smart recipients) are interesting from a theoretical point of view, since they build a link to research on herd behaviour in financial markets [34, 35]. In addition, a game-theoretic framework suggests that spammers take the role of the third party in coordination games [36], and information from spam messages creates focal points that guide investors to a social equilibrium strategy.

However, it is important to note that our study completely disregards long-term effects of stock spam. Therefore possible permanent negative consequences for micro-cap companies do not show up in this analysis at all.⁷

Finally, the mere fact that a number of people obviously follow financial advice delivered via e-mail from unknown senders, is relevant for the stability of the entire financial system. Consider the potential of spammers replacing investment hints with credible rumours about alleged solvency problems of large banks, coupled with a call to withdraw all savings. It is conceivable that the right mixture of reaction to such messages together with lacking crisis communication

⁷ While browsing through individual charts with indicated spam arrivals, the authors found some distressing cases: it appeared like stock spam systematically squeezed down some of the victim's market value. This fits also into the big picture of [17].

may actually result in a self-fulfilling prophecy, yielding bank-runs and domino effects with all their adverse consequences. Even if the probability of such co-occurrence is deemed low, such scenarios should certainly be on the agenda of crisis teams in financial supervision authorities and in cyber-terrorism defense.

4.2 Limitations and Future Work

Although our results do not leave too many puzzles and proved to be quite robust, there exists a long list of possible improvements for future research.

Probably the most salient limitation concerns the absence of a sound method to test causality. Though we made efforts to rule out the possibility that high abnormal returns are a result of the spammers' ticker selection strategy, we did not control for third variables that might affect both stock price and spammers' decision to target a particular ticker. Since basic epistemology tells us that we will never succeed in controlling for *all* possible third variables, some uncertainty will always remain and therefore we should rather think of a relationship instead of a causal link between stock spam and market indicators.

Another shortcoming of our study is the incomplete data from publicly available sources. A first step would be to collect price information for all 391 unique tickers in the SSEM data. Improving the resolution of data (down to the tick level of individual transactions) would perhaps enable us to tell the orders of spammers better apart from assumed reactions of the recipients. More consistent information on market capitalization could also be useful to weight the abnormal returns by firm size in the aggregation step. Possibly, we also face a subtle reactivity problem in the SSEM data collection. Since intra-day analyses of individual tickers are constantly published on Richardson's website [18], spammers might use this information to exclude SSEM's probe accounts from their distribution lists, and/or learn how to trick the automatic stock spam detection algorithm.

From a methodological point of view, the independence assumption about the residuals in the regression models, and particularly in the event study analysis, could be replaced by more realistic (and complicated) assumptions, possibly in a proper framework for time series analysis.

We also have in mind a number of extensions. As all spam messages are available in plain text, computer-linguistic content analyses could reveal possible clusters of origin. Moreover, the influence of specific characteristics of the message draft on its market impact could yield interesting findings on persuasive elements and response rates that are hard to obtain otherwise (because spammers usually do not disclose their response rates). Finally, a careful analysis of properties of the affected companies could help us to understand the spammers' strategy to select stocks. The determinants of spam attacks might also be of interest for CFOs of public enterprises traded on OTC markets.

To sum it all up, although we found a clear relationship in our data, we are aware that the evidence for stock spam *causing* people to buy a certain stock is not rock solid. Therefore this paper should be regarded as a first and modest step into an interesting direction.

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Appendix

Table 6. Tickers cited in stock spam (ranked by number of event days)

Rank	Ticker symbol	Company Name	Event days	Messages received
1.	VNBL.OB	VINOBLE INC	77	890
2.	CWTD.OB	CHINA WORLD TRADE	52	919
3.	NOTE.PK	NA	51	53
4.	EOGI.PK	EMERSON OIL & GAS	45	565
5.	CGKY.PK	CARNEGIE COOKE & COM	37	1320
6.	AMBD.PK	AMERICAN MOBILE DENT	37	57
7.	YPIL.PK	IPACKETS INTL INC	35	769
8.	APWL.PK	ADVANCED POWERLINE	31	240
9.	CEOA.PK	CEO AMERICA INC	31	347
10.	MOGI.PK	MONTANA OIL & GAS	29	424
11.	MWIS.OB	M-WISE INC	27	400
12.	EXTP.PK	EXTREME POKER LTD	26	84

Time period: Nov 2004 – Feb 2006

Source: [18]

Table 7. Tickers cited in stock spam (ranked by total number of messages)

Rank	Ticker symbol	Company Name	Event days	Messages received
1.	CGKY.PK	CARNEGIE COOKE & COM	37	1320
2.	CWTD.OB	CHINA WORLD TRADE	52	919
3.	VNBL.OB	VINOBLE INC	77	890
4.	YPIL.PK	IPACKETS INTL INC	35	769
5.	USTA.PK	US STARCOM INC	11	608
6.	VERY.PK	VERIDA INTERNET CORP	9	577
7.	EOGI.PK	EMERSON OIL & GAS	45	565
8.	DKDY.OB	DARK DYNAMITE INC	14	481
9.	MOGI.PK	MONTANA OIL & GAS	29	424
10.	PPTL.PK	PREMIUM PETROLM NEW	25	416
11.	MWIS.OB	M-WISE INC	27	400
12.	SCRE.OB	HUIFENG BIO-PHARM TH	21	400

Time period: Nov 2004 – Feb 2006

Source: [18]

Table 8. Estimated coefficients for effect of stock spam on traded volume

Parameter	Model			
	M1	M2	M3	M4
Terms				
Constant ($\log v_0$)	9.70 *** (0.152)	9.71 *** (0.152)	9.70 *** (0.152)	9.70 *** (0.152)
$\log \alpha$	1.15 *** (0.067)	—	—	0.94 *** (0.177)
$\log \alpha_{\text{premarket}}$	—	0.93 *** (0.102)	—	—
$\log \alpha_1$	—	—	0.76 *** (0.140)	—
$\log \alpha_2$	—	—	1.04 *** (0.190)	—
$\log \alpha_{[3,4]}$	—	—	0.92 *** (0.225)	—
$\log \alpha_{[5,8]}$	—	—	1.06 *** (0.154)	—
$\log \alpha_{[9,16]}$	—	—	1.30 *** (0.177)	—
$\log \alpha_{[17,32]}$	—	—	1.64 *** (0.191)	—
$\log \alpha_{[33,+\infty]}$	—	—	1.57 *** (0.176)	—
Trend interaction ($\log \beta_1$)	—	—	—	0.36 (0.282)
Controls				
$\log w_{\text{Tue}}$	0.05 (0.029)	0.05 (0.029)	0.05 (0.029)	0.05 (0.029)
$\log w_{\text{Wed}}$	0.04 (0.029)	0.04 (0.029)	0.04 (0.029)	0.04 (0.029)
$\log w_{\text{Thu}}$	0.02 (0.029)	0.01 (0.029)	0.02 (0.029)	0.02 (0.029)
$\log w_{\text{Fri}}$	-0.01 (0.029)	-0.02 (0.029)	-0.01 (0.029)	-0.01 (0.029)
Volume trend ($\log \beta_0$)	0.95 *** (0.033)	0.96 *** (0.033)	0.95 *** (0.033)	0.95 *** (0.033)
Summary				
Std. dev. ($\hat{\sigma}_\zeta$)	1.57	1.57	1.57	1.57
Std. dev. ($\hat{\sigma}_\epsilon$)	1.50	1.51	1.50	1.50
AIC	101810	102015	101807	101811
BIC	101884	102089	101931	101893
No. of observations (t, i)	27708	27708	27708	27708
No. of tickers i	111	111	111	111
No. of spam events	532	222	532	532

Standard errors in brackets; sig. levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9. Results of the event study analysis: estimated abnormal returns

Model	Days after event				
	0	1	2	3	4
Baseline model					
AR (in %)	1.7 *** (0.26)	-0.9 *** (0.26)	0.9 *** (0.26)	-1.1 *** (0.26)	-0.9 *** (0.26)
CAR (in %)	1.7 *** (0.26)	0.9 *** (0.36)	1.2 *** (0.45)	0.2 (0.52)	-0.7 * (0.59)
<i>n</i>	152	152	151	150	149
Models on subsets of events					
Spam before market hours only					
AR (in %)	2.1 *** (0.29)	-1.0 *** (0.29)	1.6 *** (0.30)	-1.0 *** (0.30)	-0.8 *** (0.30)
CAR (in %)	2.1 *** (0.29)	1.1 *** (0.42)	2.2 *** (0.52)	1.1 *** (0.60)	0.4 (0.67)
<i>n</i>	136	136	135	135	134
Stocks with <i>high</i> liquidity					
AR (in %)	4.3 *** (0.42)	-1.4 *** (0.42)	2.5 *** (0.42)	1.9 *** (0.43)	-1.3 *** (0.43)
CAR (in %)	4.3 *** (0.42)	2.9 *** (0.59)	5.4 *** (0.73)	7.6 *** (0.87)	6.3 *** (0.97)
<i>n</i>	41	41	41	40	40
Stocks with <i>low</i> liquidity					
AR (in %)	0.7 (0.77)	-2.2 *** (0.77)	2.5 *** (0.77)	-1.6 *** (0.77)	-0.8 * (0.77)
CAR (in %)	0.7 (0.77)	-1.5 ** (1.08)	1.0 (1.33)	-0.6 (1.53)	-1.4 (1.71)
<i>n</i>	44	44	44	44	44
Events <i>before</i> July 1st, 2005					
AR (in %)	-0.1 (0.47)	-0.7 ** (0.47)	1.3 *** (0.47)	-3.0 *** (0.47)	-2.1 *** (0.47)
CAR (in %)	-0.1 (0.47)	-0.8 * (0.67)	0.5 (0.81)	-2.5 *** (0.94)	-4.6 *** (1.05)
<i>n</i>	63	63	63	63	63
Events <i>after</i> July 1st, 2005					
AR (in %)	3.0 *** (0.50)	-1.0 *** (0.50)	0.6 * (0.51)	0.3 (0.52)	-0.0 (0.53)
CAR (in %)	3.0 *** (0.50)	2.0 *** (0.71)	1.8 *** (0.88)	2.2 *** (1.03)	2.1 *** (1.18)
<i>n</i>	89	89	88	87	86
Events with more than 10 message arrivals					
AR (in %)	7.3 *** (0.88)	-1.2 * (0.88)	1.9 *** (0.88)	-0.7 (0.88)	-0.7 (0.88)
CAR (in %)	7.3 *** (0.88)	6.1 *** (1.24)	8.0 *** (1.52)	7.3 *** (1.76)	6.6 *** (1.96)
<i>n</i>	33	33	33	33	33

Standard errors in brackets; sig. levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$